

Data Structures And Algorithm

Sem 3

Watermarked study resource

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===== SOURCE_FILE: sem 3/Data Structures And Algorithm/PYQs/Important Topics.pdf =====

CATEGORY: PYQs

STATUS: ok

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DSA:

UNIT 1:

Structure declaration, initialisation and accessing with example - 12m

Structure declaration using pointers with example - 8m

Dynamic memory allocation(all three functions) with example - 12m

Self - referential structures with example - 8m

Classification of data Structures(Types and syntax) - 8m

Abstract Data Type - 8m

Time and space complexity(Very important) - 12m

UNIT 2:

Array implementation of list and their operations: Traversing, Insertion and Deletion(algorithm and syntax) (either of two operations - 8m, all operations - 12m)

Linked list's implementation and their operations: Insertion, Deletion and Search(Algorithm and syntax) - 12m

Double linked list and their operations(Insertion and Deletion) - 8m

Josephus Problem - 8m

Polynomial and Arithmetic operations on it, write algorithm and example program(very important) - 12m

Sparse Matrix - 8m

UNIT 3:

Stack and its operations with example - 8m

Stack implementation using array with example - 8m

Stack implementation using linked list with example - 8m

Infix to postfix conversion (Very important) - 12m

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Postfix evaluation and balancing symbols - 4m or 8m

Recursion and it's example - 8m

Tower of Hanoi (Very important) - 8m

Queue implementation using linked list - 8m

Priority Queue with example - 8m

UNIT 4:

Binary Tree traversal (three types, inorder, preorder and postorder) - 12m

Binary Search Trees and its operations - 12m

AVL tree - 8m

Hash functions - 8m

Collision Handling - 8m

UNIT 5:

BFS - 8m

DFS - 8m

Prim's algorithm - 8m

Kruskal's algorithm - 8m

Dijkstra's algorithm(diagram, cost and steps) (Very important) - 12m

===== SOURCE_FILE: sem 3/Data Structures And Algorithm/PYQs/PYQ CT 2024 & 2025.url =====

CATEGORY: PYQs

STATUS: url_reference_only

URL: https://drive.google.com/drive/folders/1-A8hI-W0i_vlKHuI5q3l5XlBjb15ASQF?usp=sharing

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